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Machine Learning for Fault Detection in Wireless Sensor Networks: A Survey

Bashir Shehu and Ivan Jordanov

***Abstract*—Machine learning (ML) has become central to modern fault detection in Wireless Sensor Networks (WSNs), enabling improved robustness over traditional threshold-based schemes. However, deploying ML in WSNs remains challenging task due to the tight constraints of sensor nodes, absence of standardized datasets, and the growing need of interpretable and energy-efficient decision-making in safety-critical environments. This survey provides a structured and up-to-date review of the state of the art in ML-based fault detection for WSNs, analyzing fault taxonomies, learning paradigms, model placement strategies, dataset limitations, and evaluation practices. Particular emphasis is placed on two emerging challenges: energy efficiency, which affects the feasibility of on-node and cluster-level inference, and interpretability, which is increasingly required for transparent and trustworthy deployments. We compare lightweight and complex ML models, review intrinsic and post-hoc explanation methods, and provide a unified benchmarking perspective through dataset categorization and evaluation criteria. Finally, we highlight persistent gaps and outline a research agenda toward reliable, interpretable, and resource-aware ML frameworks for next-generation WSN fault detection.**

***Index Terms*—Energy-efficiency, explainable AI (XAI), fault detection, machine learning (ML), wireless sensor networks (WSNs).**

I. INTRODUCTION

WIRELESS sensor networks (WSNs) play a critical role in modern Internet of Things (IoT) infrastructures, supporting applications such as environmental monitoring, smart agriculture, industrial automation, and structural health assessment and others [1], [2]. These networks operate through distributed, battery-powered sensor nodes that continuously collect, process, and transmit data. However, due to harsh deployment conditions, hardware degradation, and energy constraints, sensor nodes are highly susceptible to faults that can significantly degrade data quality and system reliability [2], [3]. Ensuring timely and accurate fault detection is therefore essential for sustaining long-term WSN performance.

Machine learning (ML) has emerged as a powerful approach for detecting faults in WSNs, and with supervised models such *as Decision Trees, Naïve Bayes, Random Forests, and Support Vector Machines* has demonstrated high detection accuracy across various datasets and deployment scenarios [1], [4], [5], [6]. Despite these advances, several persistent limitations hinder the practical adoption of ML-based fault detection in real-world WSN environments. Current models often prioritize accuracy at the expense of computational load, communication cost, and explainability which are critical factors in resource-constrained, mission-critical IoT deployments [7], [8].

A key challenge lies in the lack of energy-efficient and interpretable ML models tailored to the strict limitations of WSN nodes. Models with strong performance frequently require substantial processing and memory resources or introduce communication overheads that accelerate energy depletion [8], [9], [10]. At the same time, interpretability which is essential for transparency, decision auditing, and operator trust remains underexplored in the WSN fault-detection domain, where most approaches provide limited insight into model decisions [11]. In parallel, existing evaluation practices often rely on small, short-term, or highly controlled datasets that fail to capture the dynamic and heterogeneous nature of real-world WSN deployments [3].

These gaps highlight a pressing need for a comprehensive survey that unifies the themes of energy efficiency and interpretability within ML-based fault detection for WSNs. While previous surveys have reviewed ML techniques for WSNs, none (to the best of our knowledge) have explicitly addressed the joint optimization of energy usage and interpretability as a crucial intersection for building deployable, trustworthy, and long-lived fault management systems. Furthermore, no existing work provides a structured analysis of how these dimensions interact with dataset characteristics, network constraints, and evaluation environments.

The aim of this survey is to provide a consolidated and structured overview of the state of the art in energy-efficient and interpretable ML-based fault detection for WSNs. Drawing exclusively from contemporary literature, this work: (i) analyses existing ML methods with respect to accuracy, energy usage, and interpretability; (ii) reviews the limitations of current datasets, evaluation practices, and simulation tools; and (iii) synthesizes these findings into a coherent landscape that identifies open challenges and research opportunities. By integrating perspectives from ML, embedded intelligence, and WSN design, this survey establishes a foundation for future advances in lightweight, trustworthy, and practical fault detection systems for emerging IoT applications.

II. BACKGROUND AND MOTIVATION

Machine learning–based fault detection in WSNs has progressed rapidly over the past decade, yet several foundational challenges continue to hinder practical

This work was funded in part by the Nigerian Government under the PhD Overseas Scholarship Scheme of the Petroleum Technology Development Fund (PTDF). (*Corresponding author: Bashir Shehu).*

B. Shehu is with School of Computing, University of Portsmouth, Portsmouth, P01 3HE UK (e-mail: bashir.shehu@port.ac.uk).

I. Jordanov is with School of Computing, University of Portsmouth, Portsmouth, P01 3HE UK (e-mail: ivan.jordanov@port.ac.uk).

deployment, including node energy constraints, limited interpretability of ML decisions, and unrealistic evaluation practices. Understanding the operational characteristics of WSNs, the role of ML in fault detection, and the shortcomings of current methodologies provides the necessary context for analysing and advancing energy-efficient and interpretable ML approaches. This section outlines the key background concepts and motivates the need for a focused survey of these emerging themes.

A. Wireless Sensor Networks and Fault Vulnerabilities

Wireless Sensor Networks are composed of spatially distributed sensor nodes that monitor environmental or physical conditions and forward observations toward a sink or base station [12]. These nodes are typically battery-powered, computationally constrained, and deployed in challenging environments where maintenance is infrequent or infeasible [1]. Consequently, they are susceptible to a broad spectrum of faults, including transient spikes, sensor drift, bias faults, stuck-at readings, communication failures, and hardware degradation [5], [6], [12]. If left undetected, such faults can propagate through the network and compromise application reliability, data fidelity, and energy efficiency.

Traditional rule-based or threshold-driven fault detection schemes offer computational simplicity but lack robustness to noise, contextual variations, and nonlinear sensing behaviours common in real deployments [4]. This has driven the growing adoption of ML techniques, which can capture complex relationships in sensor data, detect subtle anomalies, and adapt to evolving network conditions.

B. Machine Learning for Fault Detection in WSNs

Supervised ML methods including *Decision Trees, Naïve Bayes, Random Forests, k-Nearest Neighbours, Neural Networks, and Support Vector Machines* have demonstrated high fault-classification accuracy across a range of WSN scenarios [6], [13]. Reported accuracies frequently exceed 90% under controlled laboratory or small-scale deployments, underscoring ML's potential to improve upon conventional fault detection strategies [5], [14].

Despite these promising results, several practical limitations hinder real-world deployment. Many ML models are trained on limited or short-duration datasets that do not capture the heterogeneity and long-term dynamics of operational WSNs, weakening generalisability [1], [15]. Moreover, the computational and communication overhead associated with certain ML models often exceeds the tight energy budgets of sensor nodes, particularly when real-time inference is required [12], [16], [17]. In addition, the prevalence of opaque, black-box models reduces transparency, making it difficult for operators to understand why a particular sensor is flagged as faulty an issue of significant importance in safety-critical applications such as healthcare, structural monitoring, and industrial IoT systems.

C. Limitations of Current Evaluation Practices

A key challenge in advancing ML-driven fault detection for WSNs lies in the lack of realistic and comprehensive evaluation environments. Many studies rely on short-term laboratory datasets or simplified simulations that fail to capture real-world factors such as multi-hop routing behaviour, network congestion, node failures, energy depletion, and environmental variability [1], [17], [18]. These limitations often result in overly optimistic performance estimates that do not translate well to actual deployments.

Energy consumption which is a central concern in WSN design is also frequently overlooked in ML evaluations. Few studies quantify the energy cost of feature extraction, model inference, data communication, or decision-triggered transmissions [19]. As a result, the literature lacks standardised benchmarking protocols capable of jointly assessing accuracy, interpretability, and energy efficiency. This gap makes it difficult to compare results across studies and inhibits meaningful progress toward deployable solutions.

D. Motivation for a Survey on Energy-Efficient and Interpretable ML

Although ML-based fault detection has been extensively studied, existing surveys fall short in addressing the combined challenges of energy efficiency and interpretability two factors that fundamentally determine whether an ML model can operate on low-power sensor nodes and be trusted by human operators [1], [20], [21]. Current surveys tend to focus on fault detection broadly, on anomaly detection techniques, or on general ML applications in WSNs, but none provide a dedicated analysis of how energy-aware ML techniques intersect with interpretable modelling approaches.

Given the growing demand for transparent, resource-efficient intelligent sensing, there is a clear need to synthesise:

1) Lightweight supervised-learning models suitable for computation and energy-limited sensor nodes;
2) Interpretable ML techniques that provide clarity and trust in decision-making;
3) Evaluation practices that reflect real-world WSN constraints;
4) Insights into dataset limitations and benchmarking inconsistencies.

This survey addresses this gap by providing a systematic, WSN-specific review of energy-efficient and interpretable ML approaches for fault detection. The goal is to consolidate the fragmented literature into a coherent perspective that supports the development of robust, transparent, and practically deployable ML-based fault detection systems across the WSN application domains.

III. Taxonomy of ML-Based Fault Detection in WSNs

Machine learning approaches for fault detection in WSNs vary widely in terms of learning paradigms, target fault types, computational demands, and deployment strategies. To establish conceptual clarity and organize the diverse literature, this section presents a structured taxonomy encompassing the major dimensions along which ML-based WSN fault detection methods differ. The complete taxonomy is illustrated in Fig. 1, which summarizes these five dimensions and provides a

structured overview for the detailed analysis presented in the subsequent subsections. Fig. 1 is organized along five key dimensions: fault types, learning paradigms, model complexity, inference location, and evaluation methodology.

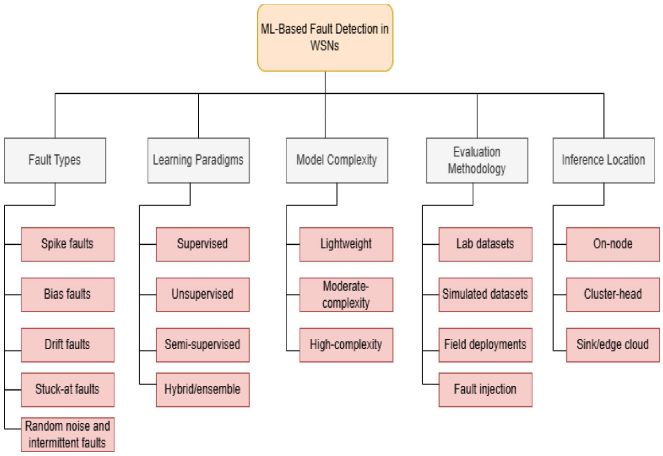


Fig. 1. Taxonomy of ML–based fault detection approaches in WSNs.

A. Fault Types and Detection Objectives

Faults in WSNs can manifest at different levels of sensing and communication, and ML models often target one or several categories. Common fault types include:

1) *Spike faults:* Sudden high-magnitude deviations induced by interference or hardware glitches [1], [22], [23], [24];
2) *Bias faults*: Consistent offset in measurement values due to ageing or calibration drift [22], [24];
3) *Drift faults*: Gradual deviation over time, typically arising from environmental changes or sensor wear [20], [22], [24], [25];
4) *Stuck-at faults*: Sensor output remains fixed despite changing conditions [1], [22], [23], [24], [26];
5) *Random noise and intermittent faults*: Sporadic, short-lived anomalies common in harsh deployments [1], [22], [23], [24], [26].

Traditional rule-based methods struggle with differentiating these faults from legitimate variations, whereas ML models learn distinguishing patterns from data. Detection objectives typically include fault classification, binary anomaly detection, or prediction-based detection (residual modelling [5], [27], [28], [29].

B. Learning Paradigms for Fault Detection

ML-based fault detection in WSNs spans several learning paradigms:

1) *Supervised learning*: Models such as *Decision Trees, SVMs, Random Forests, k-NN, NN, and Naïve Bayes* require labelled datasets and provide high detection accuracy under controlled settings [22], [23], [24], [27], [28], [29], [30];
2) *Unsupervised learning*: *Clustering* (e.g., *k-means, DBSCAN)* and dimensionality-reduction techniques (e.g., *PCA, autoencoders*) detect deviations from normal patterns without labels [26], [27], [31], [32];
3) *Semi-supervised learning*: Combines small labelled sets with larger unlabelled datasets, useful when faults are rare or expensive to annotate [14], [28], [33], [34], [35];
4) *Hybrid / ensemble learning*: Integrates multiple models to enhance robustness, though often with increased computational cost [36], [37], [38], [39].

Supervised methods dominate due to accuracy, while unsupervised/hybrid approaches offer greater scalability and adaptability.

C. Model Complexity and Resource Requirements

ML models differ widely in computation and memory requirements:

1) *Lightweight models*: *Decision Trees, Logistic Regression*, and *Naïve Bayes* offer low inference cost and suitable for on-node deployment [4], [6], [40], [41], [42];
2) *Moderate-complexity models*: *Random Forests, k-NN*, and *SVMs* offers improved accuracy but higher energy and storage demands [4], [6], [43], [44];
3) *High-complexity models*: *Neural networks, autoencoders*, and *gradient-boosted ensembles* offers high predictive power but often too computationally expensive for WSN nodes unless heavily compressed [45], [46], [47], [48], [49].

Model complexity directly influences energy consumption and inference latency, forming the basis for the energy-efficient analysis given in Section IV.

D. Evaluation Methodology and Dataset Characteristics

Different studies evaluate fault detection models using:

1) *Laboratory datasets*: Controlled environments with limited variability [50], [51];
2) *Simulated datasets*: Generated using tools such as Cooja or TOSSIM; scalable but often unrealistic [50], [52], [53];
3) *Field deployments*: Long-term multi-node studies providing realistic conditions but rarely available [50], [54];
4) *Synthetic fault injection*: Artificial bias, drift, or spike faults added to expand dataset diversity [50].

Most existing work relies on small or simplified datasets, limiting generalizability. This survey highlights the importance of realistic benchmarking (Section VI).

E. Inference Location and Deployment Strategies

WSN fault detection methods differ in where ML inference is performed:

1) *On-node inference*: Minimizes communication cost; requires lightweight, interpretable models [55], [56], [57], [58];
2) *Cluster-head inference*: Nodes offload aggregated data to a more capable intermediate node; balances computation and communication [57], [58], [59], [60], [61];

3) *Sink/edge/cloud inference*: High-capacity processing but high communication cost and energy drain [62], [63], [64], [65].

A detailed analysis of the energy implications of these deployment strategies is provided in Section IV.

IV. Energy-Efficiency Considerations In ML-enabled WSN Fault Detection

Energy consumption is the dominant constraint in WSNs, and the introduction of ML models can significantly increase computational and communication overhead. Understanding where energy is spent and how ML-based fault detection pipelines influence this distribution is essential for designing models suitable for low-power WSN deployments. Fig. 2 illustrates the major components contributing to energy consumption during ML-enabled fault detection, highlighting why lightweight, on-node inference is challenging in practical networks.

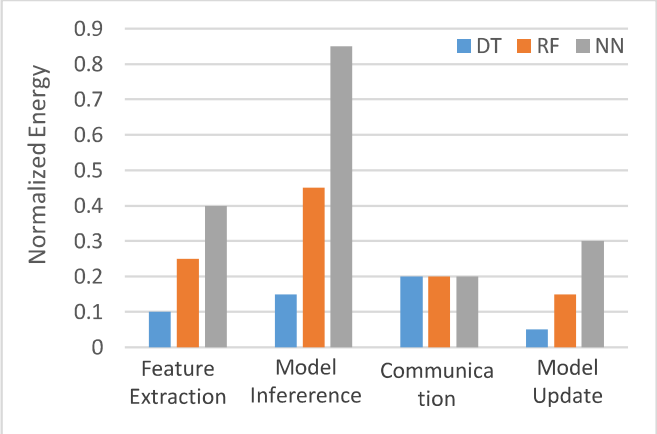


Fig. 2. Normalized energy contribution of major ML components in WSN fault detection for the three considered models: DT - *Decision Trees*; RF - *Random Forests*; and NN - *Neural Networks*.

The computational cost of ML-based fault detection varies significantly across different model architectures. Figure 2 presents a normalized comparison (0–1 scale) of the energy contributions associated with feature extraction, model inference, communication, and model update for three representative supervised models: *Decision Trees (DT), Random Forests (RF), and Neural Networks (NN)*. Rather than implying that each component accounts for a fixed percentage of total energy consumption, the figure illustrates relative energy intensities within each model category, capturing how different algorithms scale across the major processing stages.

It should be emphasized that Fig. 2 is intended as an illustrative, normalized comparison of relative energy-intensity trends across model types rather than a representation of absolute energy consumption values. The normalized scores are derived from aggregated observations reported in prior experimental and simulation-based studies, reflecting typical scaling behaviour of feature extraction, model inference, communication, and model-update costs across different supervised learning families, rather than measurements from a single hardware platform or deployment scenario. As such, the figure is designed to highlight comparative energy profiles and trade-offs between model classes, rather than to provide device-specific power estimates.

The figure shows that *Neural Networks* consistently exhibit the highest energy demand, particularly during inference, where repeated matrix multiplications lead to a normalized cost approaching 0.85. *Random Forests* demonstrate moderate inference overhead (≈0.45) but higher feature-extraction cost compared to *DTs*, since ensemble-based methods depend more heavily on multi-dimensional feature evaluation. *Decision Trees* remain the most lightweight option, with uniformly low normalized energy values across all components [1], [32], [66], [67].

Communication remains approximately constant across the three models, reflecting the fact that radio transmission cost is dominated by message size and radio duty cycle rather than model complexity [68], [69]. Model updates exhibit the lowest energy contribution for all models, aligning with the infrequent retraining cycles typically used in static-deployment WSNs [66], [70].

This normalized breakdown reinforces the importance of choosing supervised models whose computational profiles align with node-level resource constraints. It also highlights why lightweight classifiers (e.g., *DTs*) are more suitable for on-node inference, while more computationally intensive models (e.g., *RF, NN*) may need to be placed at the cluster head or gateway. These observations form the basis for the energy–accuracy trade-off discussion introduced in the subsequent sections.

A. Sources of Energy Consumption in ML-Enabled WSN Fault Detection

Energy usage in fault detection pipelines stems from:

1) *Feature extraction and pre-processing*: Techniques such as sliding windows, statistical descriptors, and frequency-domain transforms impose varying computational loads [71], [72], [73];
2) *Model inference costs*: Lightweight classifiers (e.g., *Decision Trees, Naïve Bayes*) require only a few kilobytes of memory, whereas *SVMs, k-NN*, and *Neural Networks* incur significantly higher RAM and CPU demands [74], [75], [76], [77];
3) *Communication cost due to model outputs*: Transmitting sensor readings or intermediate ML features often consumes 10× – 100× more energy than on-node computation [74], [75], [78], [79], [80]. This makes inference placement central to energy-aware design;
4) *Model update and re-training overhead*: Some methods assume periodic retraining or global synchronisation, which is rarely feasible in battery-powered deployments [71], [72], [78], [79].

Despite these well-known factors, a large proportion of ML-based WSN fault detection studies do not measure energy

usage directly, relying instead on assumptions or simulations that poorly reflect true hardware behaviour [80], [81], [82].

B. Lightweight Model Families for On-Node Deployment

Given the extreme energy constraints of sensor nodes, most real deployments require lightweight models capable of producing interpretable decisions at minimal computational cost. Models commonly considered energy-efficient include:

1) *Decision Trees*: Fast inference, small memory footprint, inherently interpretable [50], [67], [68], [79], [83];
2) *Naïve Bayes*: No iterative computation and minimal RAM usage [68], [79], [84], [85];
3) *Logistic Regression*: Low-complexity linear decision boundaries suitable for microcontrollers [84], [85].

These models typically consume millijoules per inference, making them suitable for continuous or periodic monitoring. However, they may sacrifice prediction accuracy when faults exhibit nonlinear dynamics.

In particular, their simplified decision boundaries can limit detection performance for complex or compound fault scenarios such as concurrent drift and bias faults or faults occurring under heterogeneous sensing conditions especially in dense or multi-hop WSN topologies where fault signatures may overlap [4], [6], [86].

C. Trade-offs in Moderate and High-Complexity Models

More powerful classifiers such as *Random Forests, SVMs, k-NN* and *neural networks* improve fault detection accuracy but also significantly increase energy and memory consumption [4], [6], [86]:

1) *Random Forests*: Multiple tree traversals and higher inference latency [6];
2) *k-NN*: Stores entire training dataset and prohibitive RAM usage [67], [87];
3) *SVMs*: *Support vectors* inflate computation and unsuitable for low-power nodes [4];
4) *Neural Networks/Autoencoders*: Strong detection accuracy but often exceed MCU capacity unless compressed [45], [49], [86], [88].

Despite these costs, moderate-complexity models such as *Random Forests or shallow SVMs* may be preferable in scenarios involving overlapping or non-linear fault signatures, heterogeneous sensor modalities, or clustered network topologies, where lightweight classifiers fail to reliably separate fault patterns [4], [6], [86], [89].

In such cases, the improved discriminative capability of these models can justify their deployment at cluster-heads or aggregation points, particularly when the resulting reduction in false alarms and improved fault isolation outweigh the additional energy expenditure [65], [89], [90].

These models are therefore typically deployed at cluster-heads or sink/edge devices, or are applied in hybrid frameworks (Section III-E), where only lightweight anomaly filtering occurs on-node.

D. Techniques for Reducing ML Energy Consumption

Research has explored several optimisation techniques to make ML models more energy-aware:

1) *Feature Selection & Dimensionality Reduction*: Reducing the number of features decreases both computation and communication cost [45], [91];
2) *Model Compression and Pruning*: Techniques such as quantisation, parameter pruning, and rule extraction can shrink *Neural Networks* or ensemble models to fit within node hardware limits [75], [92];
3) *Adaptive Duty Cycling with ML Assistance*: Some works integrate ML into node scheduling to reduce redundant sampling or radio transmission [93], [94];
4) *Hierarchical or Hybrid Inference Pipelines*: Simple screening on-node triggers complex analysis only when necessary, reducing total energy use [14], [36], [95], [96];
5) *Communication-Avoidant Designs*: Localised decision-making prevents frequent upstream transmissions, which often dominate energy consumption [95], [96], [97], [98].

These methods collectively highlight that energy efficiency is not merely a model-level property but an architectural design choice.

E. Limitations of Current Energy-Aware Approaches

Despite progress, the literature exhibits several persistent limitations:

1) *Lack of hardware-level energy profiling*: Many studies do not measure real MCU power draw [99], [100], [101];
2) *Over-reliance on simulations*: Energy models in simulators often fail to capture radio irregularities, environmental noise, and cross-layer behaviour [46], [102], [103];
3) *Limited consideration of interpretability:* Very few energy-efficient models also provide explainability or transparency [72], [101];
4) *Inconsistent metrics*: Studies use different units, measurement tools, and evaluation conditions, preventing direct comparison [85], [104].

As a result, end-to-end energy costs spanning sensing, feature extraction, model inference, communication, and model updates are rarely modelled holistically, leading to coarse or incomplete estimates of the true energy impact of ML-based fault detection pipelines in practical WSN deployments [46], [99], [102].

These limitations motivate the need for unified benchmarking protocols (addressed in Section VI) and explainability-driven model selection (Section V).

V. Interpretable Machine Learning for Fault Detection in WSNs

Interpretability is increasingly recognized as a critical requirement for deploying ML-based fault detection models in WSNs. Unlike cloud-based or high-resource IoT systems, WSNs often operate in safety-critical, infrastructure-dependent, or energy-limited environments where operators must understand why a model flags a node as faulty. This section reviews interpretability concepts, interpretable model families,

post-hoc explanation approaches, and the challenges of integrating them with energy efficiency.

A. Importance of Interpretability in WSN Fault Detection

Interpretability enhances operator trust, supports debugging of sensor behaviour, and enables transparent decision-making in applications such as environmental monitoring, precision agriculture, structural health monitoring, and industrial IoT [72], [105]. In WSNs, it helps:

1) Distinguish true faults from natural environmental fluctuations;
2) Reduce false alarms that waste node energy through unnecessary retransmissions or recalibration;
3) Support domain experts (e.g., environmental scientists, engineers) in validating fault origins;
4) Enable human-in-the-loop control, especially where autonomous decisions affect physical infrastructure.

Traditional black-box models such as *Neural Networks* or *SVMs* provide high accuracy but offer limited transparency, making it difficult to diagnose model failures or justify operational decisions [105], [106], [107].

B. Intrinsically Interpretable Model Families

Several lightweight supervised ML models naturally provide transparent decision logic with low memory and computation cost, making them ideal for on-node deployment in WSNs:

1) *Decision Trees*: Provide hierarchical rules that map sensor readings to fault classes; computationally inexpensive and inherently interpretable [79], [83];
2) *Rule-based Models:* Methods such as *RIPPER* or *One-R* generate human-readable decision rules that can be directly embedded into sensor node firmware [97], [108];
3) *Linear Models (e.g., Logistic Regression)*: Provide clear linear relationships between input features and fault outcomes; coefficients indicate feature importance directly [109], [110].

These models also align with energy-efficient ML design, as they require minimal RAM, CPU, and storage, making them suitable for on-node inference. However, they may struggle when fault patterns are nonlinear or when multiple interacting physical processes influence sensor readings.

C. Post-hoc Interpretability for Complex Models

When high-complexity models such as *Random Forests, autoencoders*, or *gradient-boosted ensembles* are deployed at cluster-heads or sinks, post-hoc interpretability is required to explain their decisions. Common approaches include:

1) *Feature Attribution* Methods: *SHAP (Shapley Additive Explanations)* quantifies each feature's contribution to a fault decision [72], [100], and *LIME (Local Interpretable Model-Agnostic Explanations)* approximates local decision boundaries with simple surrogate models [72], [105];
2) *Model Simplification Techniques*: Surrogate *decision trees* that approximate complex models help extract rule-based explanations for fault events [111]. Rule extraction from ensembles reduces thousands of weak learners into compact interpretable rules;
3) *Visualization-Driven Interpretability: t-SNE* or *PCA* visualizations highlight fault clusters and anomaly patterns in high-dimensional datasets [88], [112], [113].

Although these methods enhance transparency, they introduce computational overhead, making them impractical for on-node processing in most WSN scenarios. Thus, they are typically reserved for cluster-head or cloud-based inference. Table I summarizes the main intrinsically interpretable and post-hoc interpretability techniques used in WSN fault detection This table highlights a central trend: lightweight intrinsic models remain the only realistic option for on-node deployment, whereas post-hoc methods are practical only at cluster-head or sink nodes where computational resources are significantly higher.

To avoid ambiguity, it is essential to clearly distinguish between intrinsic interpretability and post-hoc explainability in WSN fault detection. Intrinsic interpretability refers to models whose decision logic is transparent by design such as *Decision Trees, rule-based classifiers, and linear models* allowing fault decisions to be directly inspected with negligible additional computation. This property makes such models fundamentally compatible with on-node deployment under strict energy, memory, and processing constraints. In contrast, post-hoc explainability applies to inherently opaque models and requires auxiliary computations to approximate or attribute model decisions. As a result, post-hoc methods are not merely less efficient alternatives but are structurally constrained to execution at cluster-heads, sinks, or edge/cloud infrastructure, reflecting a fundamental architectural limitation of WSNs rather than an implementation choice.

Furthermore, widely used post-hoc methods such as *SHAP* impose substantial computational and energy overhead due to repeated model evaluations, feature perturbations, and attribution calculations, with complexity that grows with both feature dimensionality and model size. In WSN scenarios, this overhead translates into increased energy consumption, higher latency, and, in some cases, additional communication cost when explanations must be transmitted upstream. These characteristics make *SHAP-based* explanations impractical for resource-constrained sensor nodes and reinforce their confinement to higher-tier nodes with greater computational capacity [72], [105], [114].

TABLE I
INTERPRETABLE ML TECHNIQUES FOR WSN FAULT DETECTION

Model Technique	Type	Interpretability Level	Computational Cost	Suitable Deployment	WSN Use-Cases	References

Decision Trees (DT)	Intrinsic	High (human-readable rules)	Very low	On-node	• Threshold-like anomaly identification • Multi-class bias/spike/drift classification • Rule-based binary detection • Real-time low-power inference	[1], [5], [6]
Rule-Based Classifiers (RIPPER, CART variants)	Intrinsic	Very high (logical rules)	Low	On-node / CH	• Calibration fault detection • Drift/bias tracking • Embedding domain rules • Environmental monitoring	[10], [97]
Logistic Regression (LR)	Intrinsic	Medium (feature-weight)	Very low	On-node	• Binary anomaly detection • Lightweight drift detection • Linear fault decision boundaries	[5]
Naïve Bayes (NB)	Intrinsic	Medium (probabilistic)	Very low	On-node	• Probabilistic spike detection • Handling noisy sensor data • Simple temporal anomaly detection	[55], [115]
Generalized Additive Models (GAMs)	Intrinsic	High	Moderate	Sink / CH	• Nonlinear drift modelling • Slow-changing environmental faults • Additive interpretable structure	[27]
SHAP (Shapley Values)	Post-hoc	Very high	Moderate-high	CH / Sink	• Explain RF/SVM/NN decisions • Global & local feature attribution • Debugging misclassifications	[27], [72]
LIME	Post-hoc	High	Moderate	CH /Sink	• Local instance explanations • Sparse interpretable approximations • Fault-specific reasoning	[27], [105]
Feature Importance (Gini, Permutation)	Post-hoc	Medium	Low	On-node / CH / Sink	• Global feature ranking • Lightweight model introspection • Feature reduction for efficiency	[11], [116]
Saliency / Gradient-based Maps	Post-hoc	Medium	High	Sink	• Interpreting deep models • Temporal–spatial relevance mapping • Visual justification of anomalies	[117], [118]
Surrogate Models (e.g., DT surrogate for RF)	Post-hoc	Medium-high	Moderate	CH / Sink	• Rule extraction from black-box models • Human-readable summaries of ensembles • Safety/mission assurance	[105]
Attention Mechanisms (DL)	Intrinsic - Hybrid	Medium	High	Sink	• Temporal–spatial anomaly reasoning • Highlighting critical nodes/time windows • Dense IoT stream interpretation	[32], [38]

D. *Interpretable Dimensionality Reduction and Feature Selection*

Dimensionality reduction can itself enhance interpretability by identifying the most relevant sensor features or latent patterns:

1) *Mutual information and correlation-based feature ranking*: This highlights influential variables directly [119], [120];
2) *Principal Component Analysis (PCA): PCA* identifies latent components associated with drift, bias, or environmental influences [19], [88], [112];
3) *Autoencoder-based compression*: Preserves nonlinear relationships but may sacrifice interpretability unless combined with attribution tools [14], [48], [49], [112].

Interpretable feature selection is particularly useful for energy-constrained WSNs, as reducing the feature set lowers computation and transmission cost while preserving clarity about which signals drive predictions.

E. *Challenges in Combining Interpretability and Energy Efficiency*

Despite increasing research interest, the literature shows several gaps:

1) *Lack of joint optimization*: Existing works optimize for energy or accuracy but rarely consider interpretability alongside these constraints [119], [120], [121];
2) *Post-hoc interpretability overhead*: Methods like *SHAP* or *LIME* are too heavy for typical microcontrollers and therefore unsuitable for on-node inference [88], [112];
3) *No standardized metrics*: Most studies qualitatively describe interpretability; very few provide quantitative comparison across models [21], [102];
4) *Limited real-world evaluation*: Interpretability has seldom been tested in actual deployments where fault causes must be inspected by domain experts [102].

These challenges motivate the need for integrated frameworks where energy-aware design and interpretable decision logic are co-optimized which is an issue addressed in Section VII (The Proposed Framework).

VI. Datasets and Benchmarking Practices for ML-Based WSN Fault Detection

Reliable ML-based fault detection in WSNs depends heavily on the quality, diversity, and realism of the datasets used to train and evaluate models. Unlike traditional classification tasks, WSN faults occur under complex, dynamic, and often poorly instrumented environmental conditions. As a result, dataset limitations have become a critical bottleneck for both energy-efficient model design (Section IV) and interpretable ML approaches (Section V). This section reviews common dataset categories, highlights widely used datasets,

analyses the hybrid nature of the Intel Berkeley Research Lab dataset, presents a comparative benchmarking table, and identifies persistent gaps that motivate more rigorous evaluation methodologies.

A. Overview of Dataset Types Used in WSN Fault Detection

WSN researchers typically rely on four major categories of datasets, each with unique advantages and limitations:

1) *Laboratory Datasets*: Laboratory datasets originate from controlled indoor experiments where nodes operate in stable temperature, humidity, and radio conditions. They typically have short durations, limited node diversity, and artificially introduced faults [122], [123]. Common laboratory datasets include:
   a) FIT IoT-LAB Indoor Testbeds which is a short-controlled experiment using M3, A8, and LoRa nodes, primarily containing RSSI, LQI, and temperature traces [124];
   b) IRNAS Acoustic Sensor Lab Dataset is a dataset controlled acoustic/pressure measurements with manually injected spike and bias faults [122], [123];
   c) WSN Sandbox Testbed (University-level testbeds) is a small-scale multi-node testbed used for reproducible bias, drift, and stuck-at fault studies [122], [123], [125].

   These datasets are useful for algorithm benchmarking and explainability testing (Section V), but limited in realism due to the lack of environmental variability and authentic fault progression.

2) *Field Deployment Datasets*: Field datasets are collected under natural environmental dynamics, making them crucial for validating the robustness of ML-based fault detection models. Key datasets include:
   a) GreenOrbs Forest Monitoring Dataset, a long-term multi-sensor traces collected in a forest with observable drift, environmental-induced noise, and radio irregularities [126];
   b) CitySee Urban Air-Quality Dataset, a large-scale municipal deployment with PM2.5, humidity, CO_2, and temperature, containing genuine environmental interference [127];
   c) Great Duck Island Habitat Dataset, one of the earliest WSN datasets, containing noise, humidity drift, moisture fluctuations, and intermittent node outages [42].

   These datasets contain real-world complexities essential for evaluating the interpretability of ML models (e.g., SHAP, rule extraction), as faults cannot be trivially separated from environmental fluctuations.

3) *Simulated Datasets*: Generated using WSN simulators such as Cooja, TOSSIM, or NS-3, particularly when labelled fault data are scarce. Examples include:
   a) Cooja-Sim Synthetic Fault Set, an experiment using Contiki OS nodes with controlled bias/spike/drift faults [128];
   b) TOSSIM Fault Injection Traces, link-quality and sensor value emulations used for early anomaly detection studies [27], [128];
   c) NS-3 RF Noise Simulation Dataset, RF interference and packet-loss profiles for evaluating communication-related anomalies [128].

   These datasets allow scalability testing but lack natural sensor drift, real noise, or unanticipated environmental factors. This limits the external validity of learned ML patterns.

4) *Hybrid Datasets (Laboratory + Field Components)*: Hybrid datasets combine real-world sensor measurements with synthetically injected faults, offering a balance between realism and controllable experimentation. The Intel Berkeley Research Lab dataset is the most commonly used example: it contains authentic environmental variations (e.g., day–night cycles, humidity drift, hardware ageing) but lacks ground-truth fault labels, requiring researchers to inject synthetic anomalies such as spike, bias, or drift faults. This hybrid structure preserves natural sensing behaviour while enabling supervised ML evaluation, making it more realistic than laboratory datasets yet more manageable than fully unlabelled field deployments. As a result, hybrid datasets occupy a methodological middle ground that supports supervised learning, energy-aware evaluations, and interpretability testing, where real environmental noise is essential for assessing model robustness [14], [26], [129].

B. Comparison Analysis of Widely Used WSN Fault Datasets

Table II provides a consolidated comparison of major datasets used in ML-based fault detection studies. This comparison is directly relevant to Sections III and IV, as dataset type strongly influences the realism of ML evaluation, energy consumption measurements, and the reliability of interpretability analysis. For simulated datasets, the reported duration corresponds to the configurable simulation time window rather than a fixed real-world deployment period.

TABLE II
BENCHMARK DATASETS FOR ML-BASED FAULT DETECTION IN WSNS

Dataset Name (Type)	Duration	Node Count / Density	Sensor Modalities	Fault Types Represented	Annotation Quality	Real-World Variability	Suitability for ML Tasks

FIT IoT-LAB Indoor (Lab) [124]	Hours–days	10–50	Temp, RSSI, LQI	Injected: spike, bias	High (fully labelled)	Low	Good for supervised baseline tests
IRNAS Acoustic Lab (Lab) [122], [123]	Minutes-hours	8–20	Acoustic, pressure	Injected faults	High	Low	Suitable for classification, rule extraction
WSN Sandbox Testbed (Lab) [125]	Days	10–30	Temp, humidity	Injected drift/bias	Medium–high	Low	Good for feature-selection evaluation
GreenOrbs Forest (Field) [126]	Months	100+	Temp, humidity, light	Natural drift, noise	Low (no fault labels)	High	Suitable for unsupervised, semi-supervised
CitySee Air-Quality (Field) [127]	Months	1,000+	PM2.5, CO_2, humidity	Natural noise, ageing	Low	Very high	Ideal for anomaly detection, interpretability
Great Duck Island (Field) [42]	Weeks–months	30+	Moisture, temp	Intermittent sensor dropouts	Very low	High	Suitable for robust anomaly detection
Cooja-Sim Synthetic (Simulated) [128]	Configurable (minutes-days)	10–100	Virtual sensors	Injected bias/spike	Perfect labels	Low	Suitable for algorithm stress testing
TOSSIM Fault Traces (Simulated) [27], [128]	Configurable (minutes-days)	20–50	RSSI, temp	Injected	High	Low	Benchmark for scalable ML evaluation
NS-3 RF Noise Dataset (Simulated) [128]	Configurable (minutes-days)	Variable	Radio performance	RF interference	High	Low	Communication-fault modelling
Intel Berkeley Research Lab (Hybrid) [14], [26], [129]	Months	54 nodes	Temp, humidity, light	Synthetic injection + natural drift	Medium	Medium–high	Most widely used supervised-learning dataset

C. *Limitations of Current Datasets for ML Fault Detection*

Despite a range of datasets, several critical limitations persist:

1) *Lack of standard fault annotations*: Only synthetic datasets provide explicit labels. Field datasets lack ground truth, complicating supervised training [100], [130];
2) *Limited fault diversity*: Most real datasets contain only natural faults (e.g., drift, missing data) but not engineered faults like spikes or stuck-at faults [131], [132], [133];
3) *Small-scale or short-duration data*: Few datasets exceed 100 nodes or run for more than a month [100], [134], limiting scalability assessment;
4) *Energy metrics rarely included*: Datasets seldom provide power consumption traces or MCU-level measurements, which are essential for energy-aware ML benchmarking [135], [136], [137];
5) *Inconsistent sampling rates and modalities*: Different sensors (temperature, humidity, vibration, gas, etc.) make cross-study comparison difficult [138].

Moreover, the heavy reliance on simulated environments and synthetically injected faults limits the external validity of many reported results, as learned fault patterns may not generalize to real-world deployments characterized by complex environmental dynamics and hardware ageing [100], [130], [134]. This constraint also complicates meaningful cross-study comparison, particularly when evaluating energy consumption and interpretability trade-offs under realistic operating conditions [100], [138].

These dataset limitations can further skew the reported efficacy of both energy-efficient and interpretable ML models in prior studies. In particular, evaluations based on short-duration or simulation-centric datasets often underestimate end-to-end energy costs by neglecting long-term communication overhead, adaptive duty-cycling behaviour, and hardware-level variability, leading to overly optimistic assessments of lightweight or on-node models [100], [135], [136], [138].

Similarly, interpretability analyses conducted on synthetically injected and well-separated faults may overstate explanation stability and clarity, as post-hoc or intrinsic explanation methods are rarely stress-tested against overlapping fault signatures, environmental drift, and sensor ageing observed in real deployments [72], [105], [138]. As a result, reported energy–accuracy–interpretability trade-offs may not accurately reflect deployable performance.

These limitations motivate the need for improved benchmarking and standardized evaluation protocols.

D. *Recommendations for Standardized Benchmarking in WSN ML Research*

To improve comparability and practical relevance, future research should adopt unified benchmarking practices:

1) *Multi-dataset evaluation*: Models should be validated across laboratory, field, simulated, and hybrid datasets to avoid overfitting to synthetic conditions [50], [139], [140];

2) *Energy-aware benchmarking*: Studies should report CPU cycles, RAM footprint, inference latency, and radio duty cycles using standard tools (e.g., Powertrace, Monsoon meters) [114], [138], [141];
3) *Interpretability benchmarking*: Metrics should include explanation latency, explanation stability, and perturbation sensitivity to evaluate transparent model behaviour [72], [105];
4) *Realistic fault injection standards*: Fault types and magnitudes should follow documented physical sensor models (e.g., drift rate, sensor hysteresis) [142], [143];
5) *Dataset annotation protocols:* Future deployments should incorporate automated fault logging or dual-channel sensing to provide high-quality ground truth [144], [145].

These recommendations address gaps identified in Sections III–V and prepare the foundation for the integrated framework proposed in Section VII.

VII. Toward An Integrated Framework for Energy-Efficient and Interpretable ML-Based Fault Detection in WSNs

The preceding sections highlight three converging requirements for ML-based fault detection in WSNs: (i) lightweight and energy-efficient model design, (ii) interpretable and transparent decision-making, and (iii) rigorous, realistic benchmarking across diverse dataset categories. Despite extensive research in each area individually, existing studies seldom integrate these components into a unified methodology suitable for constrained, real-world WSN deployments. This section proposes a conceptual and Methodological framework that synthesises insights from Sections III–VI and outlines a structured path for the design, optimisation, and evaluation of ML-based fault detection systems in WSNs.

Rather than presenting empirical validation or a deployment-specific case study, the framework is intended as a general design and evaluation guide that reflects current best practices and identified gaps in the literature.

A. Integrated Design Principles

The framework begins with the selection of model families aligned with WSN constraints. Lightweight, intrinsically interpretable models such as *Decision Trees, Logistic Regression,* and *Naïve Bayes* form the preferred baseline for on-node inference due to their low computational cost and transparent decision structures [41], [42], [146], [147]. Moderate and high-complexity models (e.g., *Random Forests, SVMs, autoencoders*) can be incorporated at cluster-head or sink-level processing, where energy constraints are relaxed and post-hoc interpretability tools can be applied [4], [6], [14], [72], [105]. Model selection must therefore consider a three-way trade-off between detection accuracy, energy consumption, and interpretability. This is an optimisation often overlooked in prior work.

B. Data-Driven Fault Modelling and Feature Selection

Input data quality strongly influences both model performance and interpretability. The framework emphasises the use of hybrid and field datasets, which capture natural environmental drift, sensor ageing, and heterogeneous noise patterns, ensuring that learned models generalise beyond controlled laboratory scenarios [148], [149]. Feature engineering should prioritise low-cost, physically meaningful features (e.g., moving averages, temperature gradients, humidity differentials) that facilitate both efficient computation and explainable decision paths. When dimensionality reduction is required, interpretable techniques such as *correlation ranking, mutual information*, and *PCA* should be applied with domain constraints in mind [19], [82], [89].

C. Multi-Level Inference Placement for Energy Efficiency

Building on the taxonomy in Section III and the energy analysis in Section IV, the framework adopts a hierarchical inference strategy:

1) On-node lightweight inference for rapid anomaly screening [42], [57];
2) Cluster-head inference for intermediate complexity models [59], [82];
3) Sink- or edge-level inference for deep or ensemble-based models requiring advanced interpretability tools [65], [90].

This hierarchical placement reduces communication overhead, filters benign events early, and reserves high-computation explanations or re-evaluations for centralised nodes. Such hybrid inference strategies have shown significant reductions in total network energy consumption while improving robustness to multi-type faults [65], [89], [90].

D. Interpretability as a First-Class Constraint

Interpretability is not treated as an optional post-processing step but as a core design criterion. For on-node models, interpretability is achieved through intrinsically transparent structures. For cluster-head or sink-level models, the framework incorporates post-hoc techniques such as *SHAP, LIME, surrogate trees, rule extraction*, selected based on explanation latency, stability, and compatibility with available computational resources [72], [105], [114], [150]. Explanations are integrated into the decision pipeline, enabling fault verification, model debugging, and human-in-the-loop confirmation in critical deployments.

E. Standardised Benchmarking and Evaluation Protocols

To address the inconsistencies identified in Section VI, the framework includes a set of benchmarking principles:

1) Multi-dataset evaluation across laboratory, simulated, field, and hybrid datasets [102], [140];
2) Energy profiling using standard tools (e.g., Powertrace, Monsoon Power Monitor) [83], [102];
3) Interpretability metrics such as explanation compactness, stability, and localisation accuracy [17], [72], [105];
4) Cross-layer realism, incorporating network congestion, node heterogeneity, and temporal drift [149].

These protocols ensure fair comparison of models and enable reporting of energy–accuracy–interpretability trade-offs in a standardised manner [20], [70], [83], [95], [151].

F. Summary of Framework Contribution

What distinguishes this framework from the implicit practices observed in prior studies is its explicit co-optimization of energy efficiency, interpretability, and evaluation realism within a single, hierarchical design methodology. While existing literature typically addresses these dimensions in isolation focusing on accuracy-driven model design, post-hoc interpretability, or standalone energy analysis the proposed framework integrates them as interdependent constraints that jointly shape model selection, feature engineering, inference placement, and benchmarking. In doing so, it directly responds to the gaps identified in Sections III–VI, where the absence of unified design logic, inconsistent evaluation protocols, and limited interpretability reporting were shown to hinder reproducibility and real-world applicability.

Rather than proposing a specific algorithm, this framework provides a unified methodological foundation that brings together model efficiency, interpretability, and rigorous benchmarking. It addresses long-standing gaps in the WSN literature by enabling researchers to evaluate ML models not only on accuracy but on the operational criteria essential for real deployments: computational footprint, communication overhead, energy cost, and decision transparency. This integrated approach offers a practical pathway toward deployable, trustworthy, and resource-aligned ML-based fault detection systems in WSNs.

As a concrete next step, the framework implicitly motivates the development of a community-level benchmark suite or a minimum reporting standard that jointly captures detection performance, energy consumption, and explanation cost, thereby enabling consistent and reproducible comparison of future ML-based fault detection approaches in WSNs.

As a survey-driven contribution, the framework is deliberately positioned as a reference architecture and analytical tool, with empirical validation and deployment-specific instantiation representing natural directions for future research rather than requirements of the present study.

VIII. Conclusion

Machine learning techniques have demonstrated strong potential for improving fault detection in WSNs, yet their practical deployment remains constrained by severe energy limitations, limited computational capacity, and the increasing need for transparent, interpretable decision-making. This survey has provided a unified analysis of these challenges by synthesizing research across five major dimensions: fault types and learning paradigms; model complexity; energy-efficiency constraints; interpretability methods; and dataset/benchmarking practices. Existing studies address these dimensions in isolation, but rarely integrate them resulting in models that perform well under controlled scenarios yet remain impractical for real-world WSN deployments.

By reviewing lightweight supervised models, moderate and high-complexity classifiers, intrinsic and post-hoc interpretability techniques, and the strengths and limitations of laboratory, simulated, field, and hybrid datasets, this survey highlights the critical need for joint optimization of accuracy, energy consumption, and interpretability. The integrated framework proposed in Section VII provides a cohesive methodology for designing deployable ML-based fault detection systems. It emphasizes hierarchical inference placement, interpretable feature engineering, energy-aware model selection, and standardized benchmarking protocols tailored to WSN constraints.

Despite progress in the field, several open challenges persist: the lack of ground-truth fault annotations in real-world datasets; inconsistent energy and interpretability metrics; over-reliance on synthetic fault injection; and limited cross-layer evaluation under realistic network conditions. Addressing these gaps will require coordinated efforts across ML model design, embedded systems engineering, and WSN testbed development. Ultimately, this survey lays the groundwork for developing trustworthy, resource-aligned, and practically deployable ML-based fault detection mechanisms for WSNs, advancing the state of the art toward reliable operation in diverse and dynamic IoT environments.

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